

FRB/US Macroeconomic Model in Dynare

Technical draft - July 2026

This report describes a Dynare implementation of the Federal Reserve FRB/US macroeconomic model translated from BIMETS model-definition language (MDL). The repository contains a backward-looking version and a model-consistent expectations (MCE) version. The implementation is a path-simulation replication: historical data determine equation-specific tracking residuals, and Dynare's nonlinear perfect-foresight solver evaluates counterfactual paths. Four exercises demonstrate a monetary-policy shock, the corresponding MCE experiment, propagation of model residuals, and endogenous targeting. A stochastic bootstrap is included as an optional extension.

1. Purpose and scope

FRB/US is a large-scale quarterly macroeconomic model. The goal of this repository is to provide a reproducible Dynare path-simulation version of the public BIMETS representation. The implementation does not estimate parameters or construct a posterior distribution. It preserves the translated equations and uses the observed LONGBASE path to recover add-factors that make the model reproduce the baseline.

The workflow first reconstructs the baseline, then modifies one or more add-factors or policy paths, and solves the nonlinear system again. This makes numerical errors distinguishable from differences in scenario definition.

2. Source model and translation

The source files are `source/frbus_model.txt` for the backward-looking model and `source/frbus_model_mce.txt` for the MCE model. The Python converter parses MDL identity blocks, converts time operators to Dynare timing, writes equation includes, and generates MATLAB functions for add-factor construction.

Both versions contain 284 unique endogenous variables, 293 identity blocks, and 83 listed exogenous variables. The MCE version replaces selected expectation relationships with forward-looking equations and has a maximum lead of eight quarters in the source inventory.

Selected MDL to Dynare translations

MDL operator	Dynare form
TSLAG(x,n)	$x(-n)$
TSLEAD(x,n)	$x(+n)$
TSDELTA(x)	$x - x(-1)$
TSDELTALOG(x)	$\log(x) - \log(x(-1))$
MOVAVG(x,n)	$(x + x(-1) + \dots + x(-(n-1)))/n$
LOG(x), EXP(x)	$\log(x), \exp(x)$

The seven conditional blocks in the source are retained as exact `max()` relationships. This keeps economic thresholds visible and avoids silently changing the model near a kink.

3. Tracking residuals and baseline replication

For each endogenous variable y , the translated equation is written as $y(t) = f_y(\text{current, lagged, and lead variables; exogenous inputs}) + a_y(t)$, where $a_y(t)$ is an equation-specific tracking residual. Given a data path, the add-factor function evaluates $a_y(t) = y_{\text{data}}(t) - f_y(\text{data path at } t)$. With these residuals, the perfect-foresight solve reproduces the baseline before a scenario shock is applied.

The solver initializes Dynare's original and auxiliary lag/lead variables from the same data path. This prevents zero-filled auxiliary histories from creating an artificial first-period discrepancy. The bundled CSV is a row-index-only LONGBASE export with 848 quarterly rows; row 1 is interpreted as 1962Q1 and row 848 as 2173Q4.

Statistic	Validated value
Maximum absolute error	9.50×10^{-9}
Mean absolute error	4.17×10^{-11}
Window	2040Q1 - 2045Q4

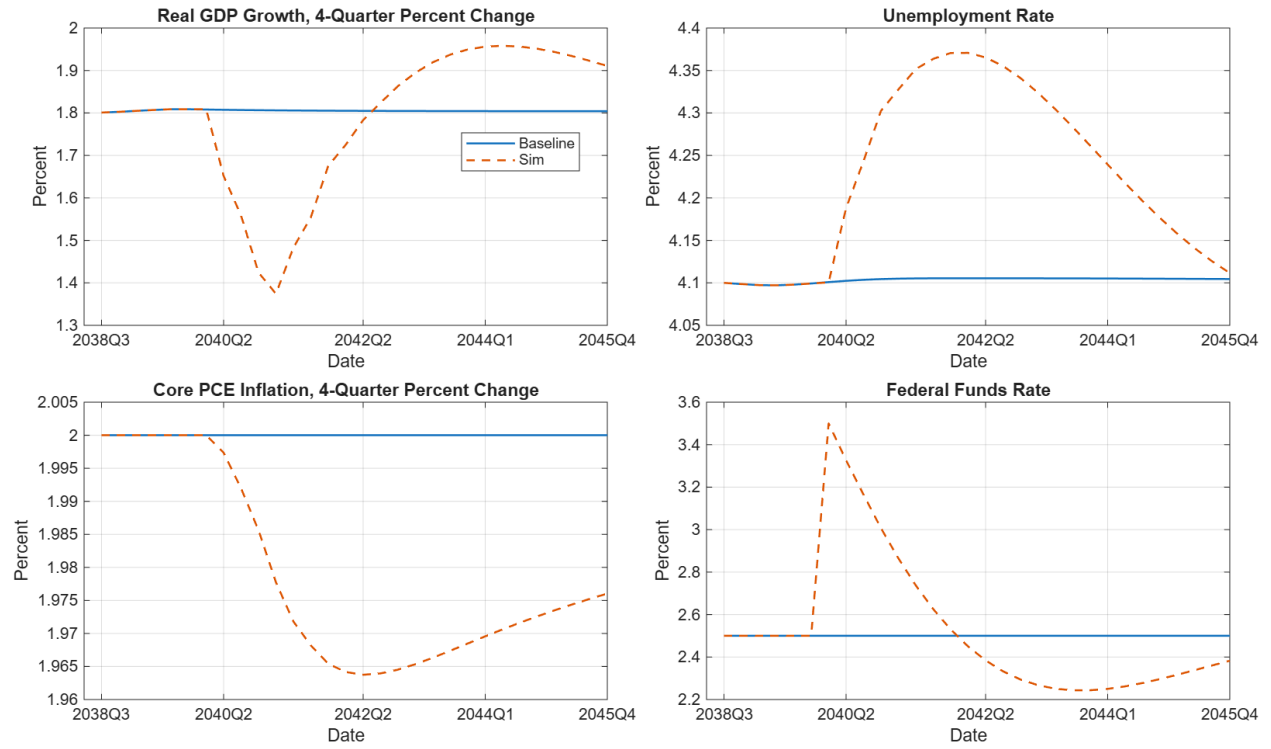
4. Dynare solution workflow

The .mod files declare the original endogenous and exogenous variables plus one add-factor for each endogenous equation. Scenario scripts load data, set policy switches, compute add-factors, initialize a finite perfect-foresight window, apply a scenario change, solve with Dynare, and export tables and pyfrbus-style charts. The MCE example requires terminal values because of forward-looking equations, so it reports a short early window.

5. Exercise 1: backward-looking monetary-policy shock

The first exercise applies a 100 basis point shock to the add-factor of `rfintay` in 2040Q1. The simulation runs through 2045Q4 with `dfpdbl = 0` and `dfpsrp = 1`. GDP growth and core PCE inflation are plotted as four-quarter changes from `xgdp` and `pcxfe`; unemployment and the federal funds rate are shown in levels.

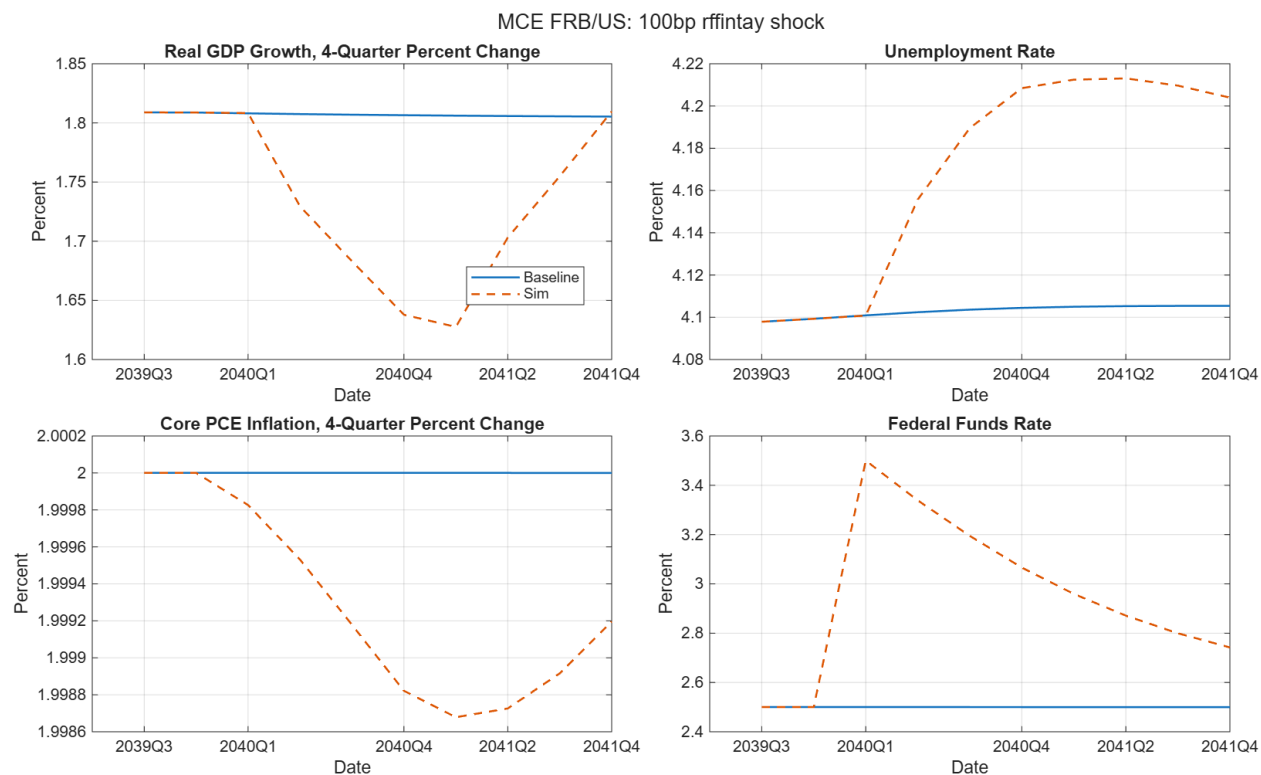
Backward-looking FRB/US: 100bp rffintay shock



Backward-looking model: 100 bp monetary-policy shock.

6. Exercise 2: model-consistent expectations shock

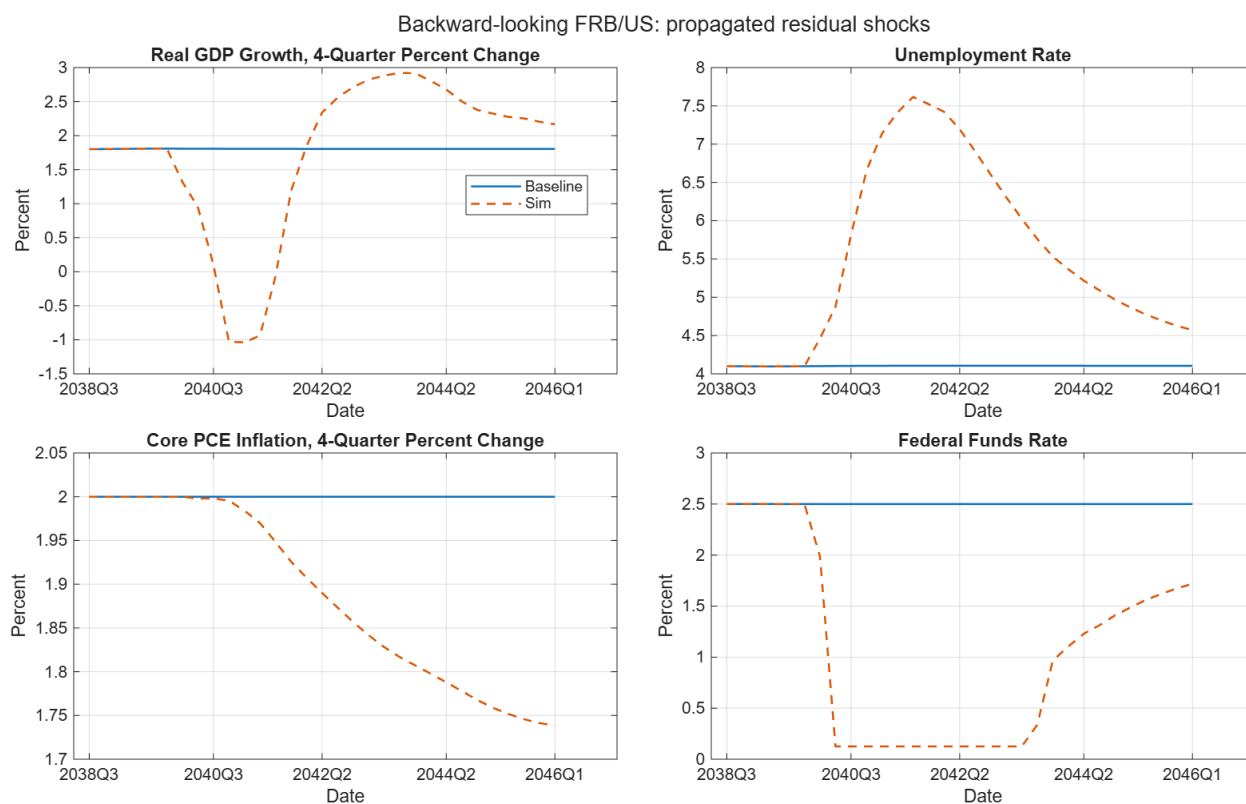
The second exercise applies the same policy shock to the MCE representation. It uses 2040Q1 - 2041Q4, sets $drstar = 0$ initially, and turns $drstar$ on from 2041Q1. The shorter window demonstrates the forward-looking model and its terminal-value requirement.



MCE model: short-horizon 100 bp monetary-policy shock.

7. Exercise 3: error propagation

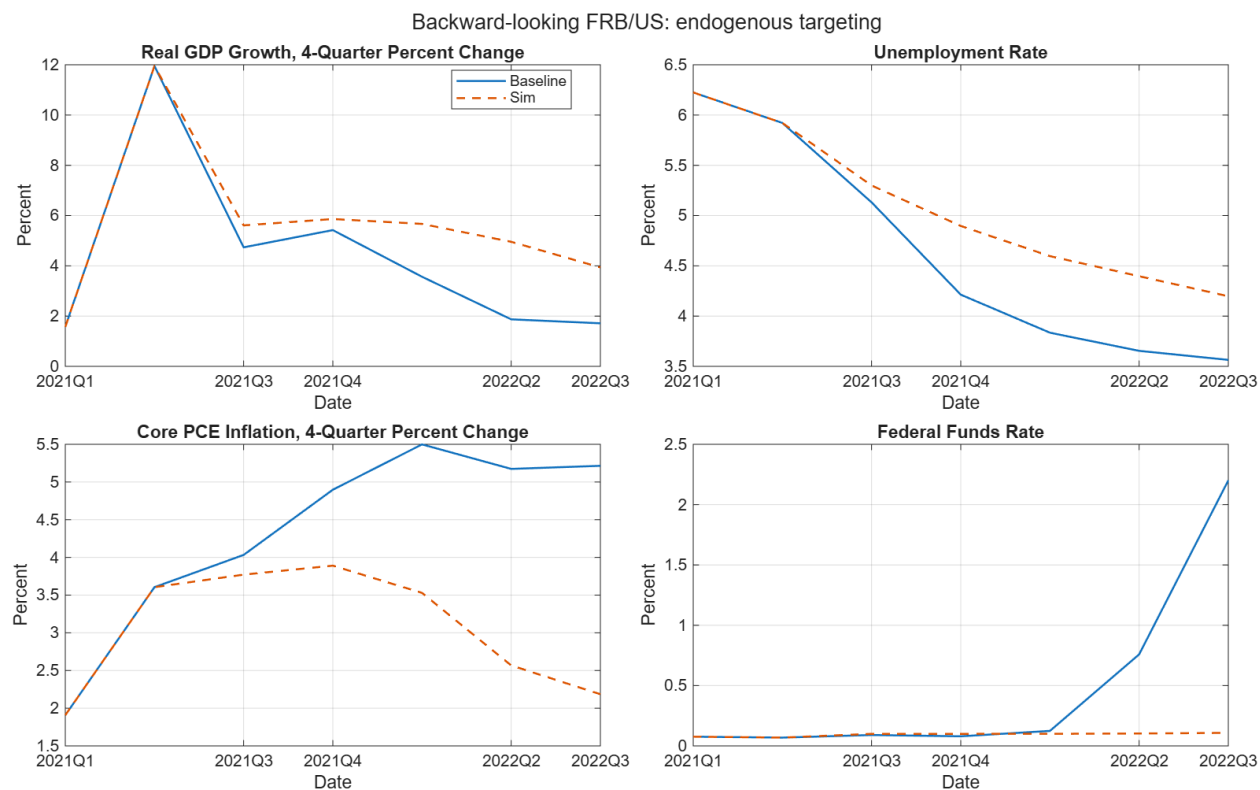
The third exercise preserves the policy configuration and applies a persistent residual shock. The shock is ramped into the model through continuation steps so that each nonlinear solve starts from the previous converged path. This is useful for a large system with threshold equations and does not alter the final scenario definition.



Backward-looking model: propagated residual shock.

8. Exercise 4: endogenous targeting

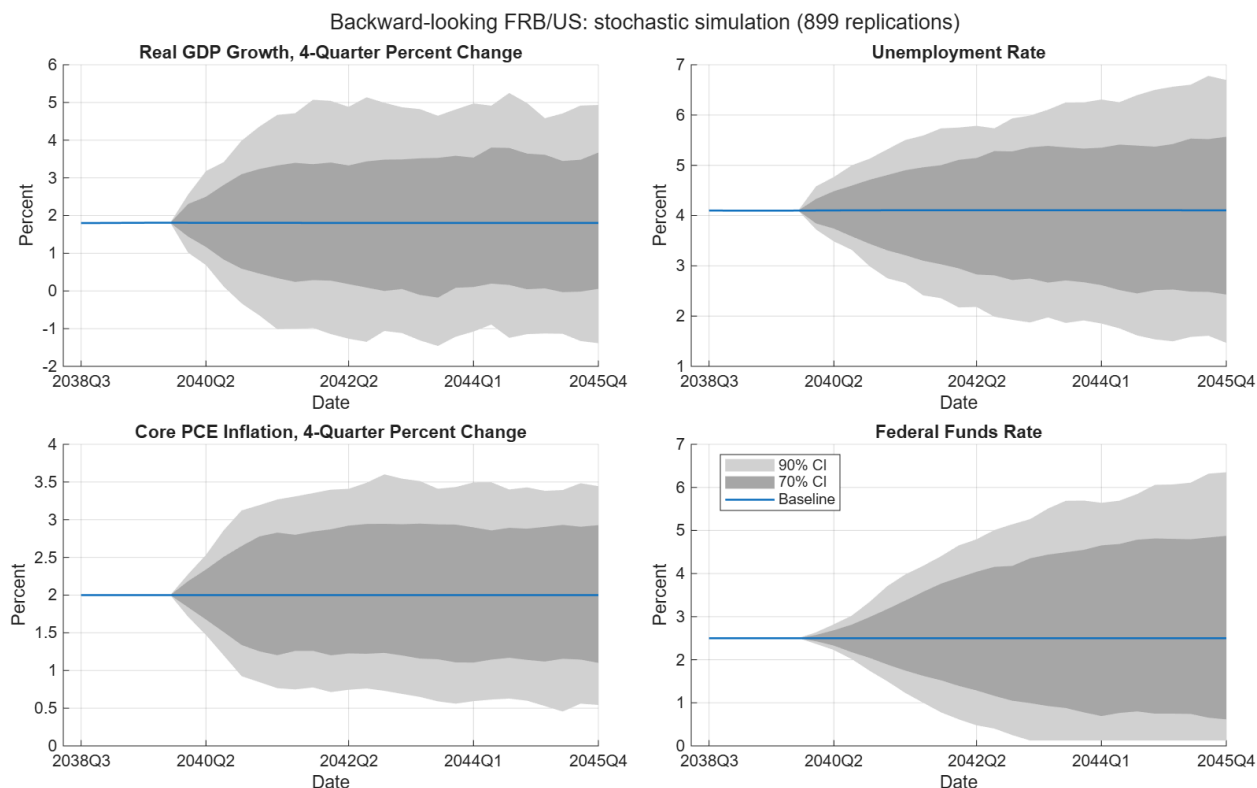
The fourth exercise implements an external Newton controller. It adjusts five add-factor instruments - *eco*, *lhp*, *picxfe*, *rff*, and *rg10p* - to match five target paths - *xgdp*, *lur*, *picxfe*, *rff*, and *rg10* - from 2021Q3 through 2022Q3. The controller uses finite differences, damping, and repeated perfect-foresight solves.



Backward-looking model: endogenous targeting exercise.

9. Optional stochastic extension

The repository also includes a serial bootstrap-style stochastic simulation. It samples historical quarters for 64 listed stochastic residuals, adds the sampled residuals to future tracking paths, and solves each draw as a deterministic perfect-foresight problem. FRBUS_NREPL controls the number of successful replications; the default is 1000 and ten replications are suitable for a smoke test.



10. Reproducibility and limitations

The README gives commands for running MATLAB, regenerating translated files, running Python tests, and compiling the LaTeX source. Dynare preprocessing creates model-specific packages under `dynare/`; those generated directories are ignored by Git and regenerated on demand.

The main limitations are the use of exact `max()` threshold equations, the cost of repeated nonlinear solves, and the short terminal horizon of the included MCE demonstration. The source equations and coefficients are not re-estimated. Comparisons with BIMETS or `pyfrbus` should distinguish model translation, data calendar alignment, add-factor construction, solver tolerances, and charting conventions.